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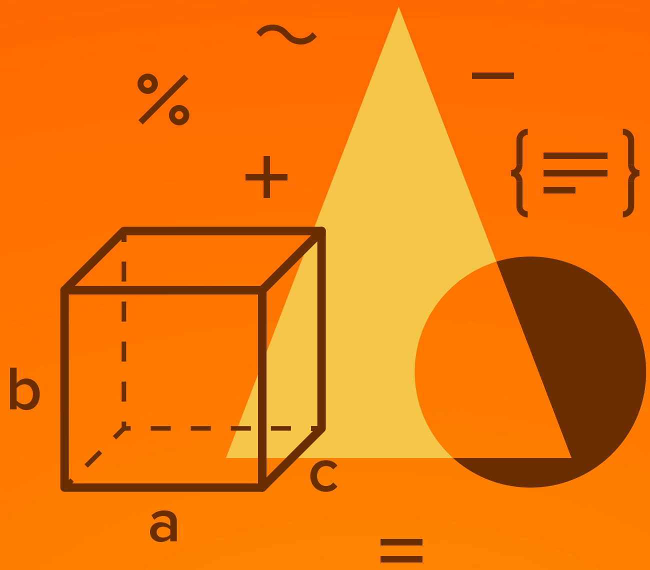


MATHEMATICS

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Complex Number

English Medium

EXERCISE (O-1)

Straight Objective Type

1. If $z + z^3 = 0$ then which of the following must be true on the complex plane?

(A) $\text{Re}(z) < 0$ (B) $\text{Re}(z) = 0$ (C) $\text{Im}(z) = 0$ (D) $z^4 = 1$

CN0029

2. Number of integral values of n for which the quantity $(n + i)^4$ where $i^2 = -1$, is an integer is

(A) 1 (B) 2 (C) 3 (D) 4

CN0030

3. Let $i = \sqrt{-1}$. The product of the real part of the roots of $z^2 - z = 5 - 5i$ is

(A) – 25 (B) – 6 (C) – 5 (D) 25

CN0031

4. There is only one way to choose real numbers M and N such that when the polynomial $5x^4 + 4x^3 + 3x^2 + Mx + N$ is divided by the polynomial $x^2 + 1$, the remainder is 0. If M and N assume these unique values, then M – N is

(A) – 6 (B) – 2 (C) 6 (D) 2

CN0032

5. Number of complex numbers z satisfying $z^3 = \bar{z}$ is

(A) 1 (B) 2 (C) 4 (D) 5

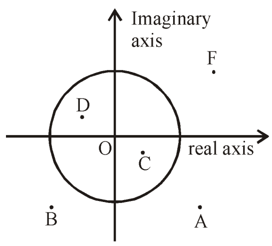
CN0036

6. Let $z = 9 + bi$ where b is non zero real and $i^2 = -1$. If the imaginary part of z^2 and z^3 are equal, then b^2 equals

(A) 261 (B) 225 (C) 125 (D) 361

CN0037

7. The diagram shows several numbers in the complex plane. The circle is the unit circle centered at the origin. One of these numbers is the reciprocal of F, which is



(A) A (B) B

(C) C (D) D

CN0039

8. If $z = x + iy$ & $\omega = \dfrac{1 - iz}{z - i}$ then $|\omega| = 1$ implies that, in the complex plane

(A) z lies on the imaginary axis (B) z lies on the real axis

(C) z lies on the unit circle (D) none

CN0040

9. On the complex plane locus of a point z satisfying the inequality $2 \le |z-1| < 3$ denotes

(A) region between the concentric circles of radii 3 and 1 centered at (1, 0)

(B) region between the concentric circles of radii 3 and 2 centered at (1, 0) excluding the inner and outer boundaries.

(C) region between the concentric circles of radii 3 and 2 centered at (1, 0) including the inner and outer boundaries.

(D) region between the concentric circles of radii 3 and 2 centered at (1, 0) including the inner boundary and excluding the outer boundary.

CN0041

10. The complex number z satisfies $z + |z| = 2 + 8i$. The value of $|z|$ is

(A) 10 (B) 13 (C) 17 (D) 23

CN0042

11. Let $Z_1 = (8+i)\sin\theta + (7+4i)\cos\theta$ and $Z_2 = (1+8i)\sin\theta + (4+7i)\cos\theta$ are two complex numbers. If $Z_1 \cdot Z_2 = a + ib$ where $a, b \in R$ then the largest value of $(a+b)$ $\forall\ \theta \in R$, is

(A) 75 (B) 100 (C) 125 (D) 130

CN0043

12. The locus of z, for $\arg z = -\pi/3$ is

(A) same as the locus of z for $\arg z = 2\pi/3$

(B) same as the locus of z for $\arg z = \pi/3$

(C) the part of the straight line $\sqrt{3}x + y = 0$ with $(y < 0, x > 0)$

(D) the part of the straight line $\sqrt{3}x + y = 0$ with $(y > 0, x < 0)$

CN0044

13. If z_1 & $\bar{z}_1$ represent adjacent vertices of a regular polygon of n sides with centre at the origin & if $\dfrac{\operatorname{Im} z_1}{\operatorname{Re} z_1} = \sqrt{2}-1$ then the value of n is equal to :

(A) 8 (B) 12 (C) 16 (D) 24

CN0045

14. For $Z_1 = \sqrt[6]{\dfrac{1-i}{1+i\sqrt{3}}}$; $Z_2 = \sqrt[6]{\dfrac{1-i}{\sqrt{3}+i}}$; $Z_3 = \sqrt[6]{\dfrac{1+i}{\sqrt{3}-i}}$ which of the following holds good?

(A) $\sum |Z_1|^2 = \dfrac{3}{2}$

(B) $|Z_1|^4 + |Z_2|^4 = |Z_3|^{-8}$

(C) $\sum |Z_1|^3 + |Z_2|^3 = |Z_3|^{-6}$

(D) $|Z_1|^4 + |Z_2|^4 = |Z_3|^8$

CN0047

15. If z is a complex number satisfying the equation $|z+i|+|z-i|=8$, on the complex plane then maximum value of $|z|$ is

(A) 2 (B) 4 (C) 6 (D) 8

CN0050

16. Let Z be a complex number satisfying the equation $(Z^3+3)^2=-16$ then $|Z|$ has the value equal to

(A) $5^{1/2}$ (B) $5^{1/3}$ (C) $5^{2/3}$ (D) 5

CN0052

17. If z_1, z_2, z_3 are 3 distinct complex numbers such that $\dfrac{3}{|z_2-z_3|}=\dfrac{4}{|z_3-z_1|}=\dfrac{5}{|z_1-z_2|}$, then the value of $\dfrac{9}{z_2-z_3}+\dfrac{16}{z_3-z_1}+\dfrac{25}{z_1-z_2}$ equals

(A) 0 (B) 3 (C) 4 (D) 5

CN0053

18. Consider two complex numbers α and β as $\alpha=\left(\dfrac{a+bi}{a-bi}\right)^2+\left(\dfrac{a-bi}{a+bi}\right)^2$, where $a, b \in R$ and $\beta=\dfrac{z-1}{z+1}$, $z \neq \pm 1$ where $|z|=1$, then

(A) Both α and β are purely real

(B) Both α and β are purely imaginary

(C) α is purely real and β is purely imaginary

(D) β is purely real and α is purely imaginary

CN0055

19. Let Z is complex satisfying the equation $z^2-(3+i)z+m+2i=0$, where $m \in R$. Suppose the equation has a real root. The additive inverse of non real root, is

(A) $1-i$ (B) $1+i$ (C) $-1-i$ (D) -2

CN0056

20. The minimum value of $|z-1+2i|+|4i-3-z|$ is

(A) $\sqrt{5}$ (B) 5 (C) $2\sqrt{13}$ (D) $\sqrt{15}$

CN0057

21. Identify the incorrect statement.

(A) no non zero complex number z satisfies the equation, $\bar{z} = -4z$

(B) $\bar{z} = z$ implies that z is purely real

(C) If z is purely imaginary, then $\bar{z} + z = 0$.

(D) if z_1, z_2 are the roots of the quadratic equation $az^2 + bz + c = 0$ such that $\operatorname{Im}(z_1 z_2) \neq 0$ then a,b,c must be real numbers .

CN0060

22. $z_1 = \dfrac{a}{1-i}$; $z_2 = \dfrac{b}{2+i}$; $z_3 = a - bi$ for $a, b \in R$ if $z_1 - z_2 = 1$ then the centroid of the triangle formed by the points z_1, z_2, z_3 in the argand's plane is given by

(A) $\dfrac{1}{9}(1+7i)$ (B) $\dfrac{1}{3}(1+7i)$ (C) $\dfrac{1}{3}(1-3i)$ (D) $\dfrac{1}{9}(1-3i)$

CN0062

23. Number of complex numbers z such that $|z| = 1$ and $\left|\dfrac{z}{\bar{z}} + \dfrac{\bar{z}}{z}\right| = 1$ is

(A) 4 (B) 6 (C) 8 (D) more than 8

CN0064

24. If z is a complex number satisfying the equation $|z - (1+i)|^2 = 2$ and $\omega = \dfrac{2}{z}$, then the locus traced by 'ω' in the complex plane is

(A) $x - y - 1 = 0$ (B) $x + y - 1 = 0$ (C) $x - y + 1 = 0$ (D) $x + y + 1 = 0$

CN0065

25. A particle starts from a point $z_0 = 1 + i$, where $i = \sqrt{-1}$. It moves horizontally away from origin by 2 units and then vertically away from origin by 3 units to reach a point z_1. From z_1 particle moves $\sqrt{5}$ units in the direction of $2\hat{i} + \hat{j}$ and then it moves through an angle of $\operatorname{cosec}^{-1}\sqrt{2}$ in anticlockwise direction of a circle with centre at origin to reach a point z_2. The arg z_2 is given by

(A) $\sec^{-1}2$ (B) $\cot^{-1}0$ (C) $\sin^{-1}\left(\dfrac{\sqrt{3}-1}{2\sqrt{2}}\right)$ (D) $\cos^{-1}\left(\dfrac{-1}{2}\right)$

CN0068

26. Given $z_p = \cos\left(\dfrac{\pi}{2^P}\right) + i\sin\left(\dfrac{\pi}{2^P}\right)$, then $\lim\limits_{n\to\infty}(z_1 z_2 z_3 \ldots z_n) =$

(A) 1 (B) -1 (C) i (D) $-i$

CN0071

27. If $|z| = 1$ and $|\omega - 1| = 1$ where $z, \omega \in C$, then the largest set of values of $|2z - 1|^2 + |2\omega - 1|^2$ equals

(A) [1, 9] (B) [2, 6] (C) [2, 12] (D) [2, 18]

CN0073

28. If $\operatorname{Arg}(z + a) = \frac{\pi}{6}$ and $\operatorname{Arg}(z - a) = \frac{2\pi}{3}$; $a \in R^+$, then

(A) z is independent of a

(B) $|a| = |z + a|$

(C) $z = a \operatorname{Cis} \frac{\pi}{6}$

(D) $z = a \operatorname{Cis} \frac{\pi}{3}$

CN0074

29. If z_1, z_2, z_3 are the vertices of the Δ ABC on the complex plane which are also the roots of the equation, $z^3 - 3\alpha z^2 + 3\beta z + x = 0$, then the condition for the Δ ABC to be equilateral triangle is

(A) $\alpha^2 = \beta$ (B) $\alpha = \beta^2$ (C) $\alpha^2 = 3\beta$ (D) $\alpha = 3\beta^2$

CN0075

30. The locus represented by the equation, $|z - 1| + |z + 1| = 2$ is :

(A) an ellipse with focii (1 , 0) ; (– 1 , 0)

(B) one of the family of circles passing through the points of intersection of the circles $|z - 1| = 1$ & $|z + 1| = 1$

(C) the radical axis of the circles $|z - 1| = 1$ and $|z + 1| = 1$

(D) the portion of the real axis between the points (1 , 0) ; (– 1 , 0) including both.

CN0076

EXERCISE (O-2)

Multiple Correct Answer Type

1. If z is a complex number which simultaneously satisfies the equations $3|z-12| = 5|z-8i|$ and $|z-4| = |z-8|$ then the Im(z) can be

(A) 15 (B) 16 (C) 17 (D) 8

CN0097

2. Let z_1, z_2, z_3 be non-zero complex numbers satisfying the equation $z^4 = iz$.
Which of the following statement(s) is/are correct?

(A) The complex number having least positive argument is $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$.

(B) $\sum_{k=1}^{3} \text{Amp}(z_k) = \frac{\pi}{2}$

(C) Centroid of the triangle formed by z_1, z_2 and z_3 is $\left(\frac{1}{\sqrt{3}}, \frac{-1}{3}\right)$

(D) Area of triangle formed by z_1, z_2 and z_3 is $\frac{3\sqrt{3}}{2}$

CN0098

3. The locii of a point P(z) in the complex plane satisfying the $\left|z + \frac{1}{z}\right| = 2$ are two circles C_1 and C_2. These circles

(A) have centres on real axis. (B) cut each other orthogonally.

(C) are congruent (D) have exactly two common tangents.

CN0100

4. Let A and B be two distinct points denoting the complex numbers α and β respectively. A complex number z lies between A and B where $z \neq \alpha$, $z \neq \beta$. Which of the following relation(s) hold good?

(A) $|\alpha - z| + |z - \beta| = |\alpha - \beta|$

(B) $\exists$ a positive real number 't' such that $z = (1-t)\alpha + t\beta$

(C) $\begin{vmatrix} z-\alpha & \bar{z}-\bar{\alpha} \\ \beta-\alpha & \bar{\beta}-\bar{\alpha} \end{vmatrix} = 0$

(D) $\begin{vmatrix} z & \bar{z} & 1 \\ \alpha & \bar{\alpha} & 1 \\ \beta & \bar{\beta} & 1 \end{vmatrix} = 0$

CN0101

5. Equation of a straight line on the complex plane passing through a point P denoting the complex number α and perpendicular to the vector $\overrightarrow{OP}$ where 'O' is the origin can be written as

(A) $\text{Im}\left(\frac{z-\alpha}{\alpha}\right) = 0$

(B) $\text{Re}\left(\frac{z-\alpha}{\alpha}\right) = 0$

(C) $\text{Re}(\bar{\alpha} z) = 0$

(D) $\bar{\alpha} z + \alpha\bar{z} - 2|\alpha|^2 = 0$

CN0104

6. If $\alpha_1, \alpha_2, \alpha_3, \ldots\ldots., \alpha_{n-1}$ are the imaginary n^{th} roots of unity then the product $\prod_{r=1}^{n-1}(i-\alpha_r)$ (where $i=\sqrt{-1}$) can take the value equal to

(A) 0 (B) 1 (C) i (D) (1 + i)

CN0105

7. If the expression $(1 + ir)^3$ is of the form of $s(1 + i)$ for some real 's' where 'r' is also real and $i = \sqrt{-1}$, then the value of 'r' can be

(A) $\cot\frac{\pi}{8}$ (B) $\sec\pi$ (C) $\tan\frac{\pi}{12}$ (D) $\tan\frac{5\pi}{12}$

CN0106

8. Locus of all points z in argand plane which satisfy $|z^2 + 1| = |z^2 - 1|$ is -

(A) same as $\text{Re}(z) = 0$ (B) same as $\text{Re}(z^2) = 0$

(C) a pair of straight lines (D) a circle with unit radius

CN0108

9. If the biquadratic $x^4 + ax^3 + bx^2 + cx + d = 0$ $(a, b, c, d \in R)$ has 4 non real roots, two with sum $3 + 4i$ and the other two with product $13 + i$.

(A) $b = 51$ (B) $a = -6$ (C) $c = -70$ (D) $d = 170$ **CN0197**

10. The quadratic equation $z^2 + (p + ip')z + q + iq' = 0$; where p, p', q, q' are all real.

(A) if the equation has one real root then $q'^2 - pp'q' + qp'^2 = 0$. **CN0198**

(B) if the equation has two equal roots then $pp' = 2q'$.

(C) if the equation has two equal roots then $p^2 - p'^2 = 4q$

(D) if the equation has one real root then $p'^2 - pp'q' + q'^2 = 0$.

11. The value of $i^n + i^{-n}$, for $i = \sqrt{-1}$ and $n \in I$ is : **CN0199**

(A) $\frac{2^n}{(1-i)^{2n}}+\frac{(1+i)^{2n}}{2^n}$ (B) $\frac{(1+i)^{2n}}{2^n}+\frac{(1-i)^{2n}}{2^n}$

(C) $\frac{(1+i)^{2n}}{2^n}+\frac{2^n}{(1-i)^{2n}}$ (D) $\frac{2^n}{(1+i)^{2n}}+\frac{2^n}{(1-i)^{2n}}$

12. If amp $(z_1z_2) = 0$ and $|z_1| = |z_2| = 1$, then **CN0200**

(A) $z_1 + z_2 = 0$ (B) $z_1z_2 = 1$ (C) $z_1 = \bar{z}_2$ (D) $z_1 = z_2$

13. Let z_1 and z_2 are two complex numbers such that $(1 - i)z_1 = 2z_2$ and $\arg(z_1z_2) = \frac{\pi}{2}$, then $\arg(z_2)$ is equal to **CN0201**

(A) $3\pi/8$ (B) $\pi/8$ (C) $5\pi/8$ (D) $-7\pi/8$

14. a, b, c are real numbers in the polynomial, $P(z) = 2z^4 + az^3 + bz^2 + cz + 3$. If two roots of the equation $P(z) = 0$ are 2 and i. Then which of the following are true. **CN0202**

(A) $a = -\frac{11}{2}$ (B) $b = 5$ (C) $c = -\frac{11}{2}$ (D) $a = -11$

15. If $Z = \frac{(1+i)(1+2i)(1+3i)......(1+ni)}{(1-i)(2-i)(3-i)......(n-i)}$, $n \in N$ then principal argument of Z can be

(A) 0 (B) $\frac{\pi}{2}$ (C) $-\frac{\pi}{2}$ (D) π **CN0203**

16. For complex numbers z and w, if $|z|^2w - |w|^2 z = z - w$. Which of the following can be true :

(A) $z = w$ (B) $z\bar{w} = 1$ (C) $z = w + 2$ (D) $\bar{z}w = 1$

CN0204

17. The sides of a triangle ABC (right angled at B) are equimultiples of three consecutive natural numbers such that perimeter of the triangle is 24 units. If z_1, z_2 & z_3 represent affixes of vertices A, B & C respectively and $B(z_2)$ is given by 6 + i.0 and AB coincide with real axis, then -

(A) inradius of the Δ is 2 (B) $|z_1|_{max} = 12$ **CN0205**

(C) circumcircle of the Δ is $\left|z - \frac{z_1 + z_3}{2}\right| = 10$ (D) $|z_1|_{min} = 6$

18. If sinA + sinB + sinC = cosA + cosB + cosC = 0, then - **CN0206**

(A) $\sin(A+B+C) = \frac{1}{3}(\sin 3A + \sin 3B + \sin 3C)$ (B) $\cos(A+B+C) = \frac{1}{3}(\cos 3A + \cos 3B + \cos 3C)$

(C) $\cos(A-B) + \cos(B-C) + \cos(C-A) = -\frac{3}{2}$ (D) $\cos 2A + \cos 2B + \cos 2C = \sin 2A + \sin 2B + \sin 2C$

19. If roots of $az^2 + az + 1 = 0$ are purely imaginary, where $a = \cos\theta + i\sin\theta$, $i = \sqrt{-1}$ then :-

(A) $\cos\theta = 2\sin\left(\frac{\pi}{10}\right)$ (B) $z = \pm i\sqrt{2\sin\left(\frac{\pi}{10}\right)}$ (C) $z = \pm\sqrt{2\sin\left(\frac{\pi}{10}\right)}$ (D) $\sin\theta = 2\sin\left(\frac{\pi}{10}\right)$

CN0207

20. Let $a = 1 + e^{i\pi} + e^{2i\pi} + e^{3i\pi} + ...$ to ∞ and $b = 1 + e^{-i\pi} + e^{-2i\pi} + e^{-3i\pi} + ...$ to ∞ then :

(A) a = b (B) a = 1/2 (C) b = 1/2 (D) none of these

CN0208

21. If z satisfies $|z + 1| < |z - 2|$ then $\omega = 3z + 2 + i$ satisfies : **CN0209**

(A) $|\omega + 1| < |\omega - 8|$ (B) $|\omega + 1 + i| < |\omega - 8 + i|$

(C) $\text{Re}\left(\frac{1}{2\omega - 7}\right) > 0$ (D) $|\omega + 5| < |\omega - 4|$

22. The equation, $z\bar{z} + a\bar{z} + \bar{a}z + b = 0$, $b \in R$, represents a circle if : **CN0210**

(A) $|a|^2 = b$ (B) $|a|^2 < b$ (C) $|a|^2 > b$ (D) none of these

23. If $z^3 - iz^2 - 2iz - 2 = 0$ then z can be equal to : **CN0211**

(A) $1 - i$ (B) i (C) $1 + i$ (D) $-1 - i$

24. A complex number z satisfying the equation, $\log_{14}(13 + |z^2 - 4i|) + \log_{196}\dfrac{1}{\left(13 + |z^2 + 4i|\right)^2} = 0$

(A) can be purely real (B) can be purely imaginary **CN0212**

(C) can be imaginary (D) must be real or purely imaginary.

25. Let z_1 and z_2 be two distinct complex numbers and let $z = (1 - t)z_1 + tz_2$ for some real number t with $0 < t < 1$. If Arg(w) denotes the principal argument of a nonzero complex number w, then

(A) $|z - z_1| + |z - z_2| = |z_1 - z_2|$ (B) $\text{Arg}(z - z_1) = \text{Arg}(z - z_2)$

(C) $\begin{vmatrix} z - z_1 & \bar{z} - \bar{z}_1 \\ z_2 - z_1 & \bar{z}_2 - \bar{z}_1 \end{vmatrix} = 0$ (D) $\text{Arg}(z - z_1) = \text{Arg}(z_2 - z_1)$

CN0164

EXERCISE (O-3)

Linked Comprehension Type

Paragraph for question nos. 1 to 3

Consider a complex number $w = \dfrac{z-i}{2z+1}$, where $z = x + iy$ and $x, y \in R$.

1. If the complex number w is purely imaginary then locus of z is -

(A) a straight line

(B) a circle with centre $\left(-\dfrac{1}{4}, \dfrac{1}{2}\right)$ and radius $\dfrac{\sqrt{5}}{4}$.

(C) a circle with centre $\left(\dfrac{1}{4}, -\dfrac{1}{2}\right)$ and passing through origin.

(D) neither a circle nor a straight line.

CN0028

2. If the complex number w is purely real then locus of z is

(A) a straight line passing through origin

(B) a straight line with gradient 3 and y intercept (–1)

(C) a straight line with gradient 2 and y intercept 1.

(D) none

CN0028

3. If $|w| = 1$ then the locus of P(z) is

(A) a point circle (B) an imaginary circle

(C) a real circle (D) not a circle.

CN0028

Paragraph for question nos. 4 to 5

ABCD is a rhombus. Its diagonals AC and BD intersect at the point M and satisfy BD = 2AC. Let the points D and M represent complex numbers 1 + i and 2 – i respecetively.

If θ is arbitary real, then $z = re^{i\theta}$, $R_1 \le r \le R_2$ lies in annular region formed by concentric circles $|z| = R_1$, $|z| = R_2$.

4. A possible representation of point A is **CN0213**

(A) $3-\dfrac{i}{2}$ (B) $1-\dfrac{3i}{2}$ (C) $1+\dfrac{3}{2}i$ (D) $3-\dfrac{3}{2}i$

5. If z is any point on segment DM then $w = e^{iz}$ lies in annular region formed by concentric circles.

(A) $|w|_{min} = 1$, $|w|_{max} = 2$ (B) $|w|_{min} = \dfrac{1}{e}$, $|w|_{max} = e$ **CN0214**

(C) $|w|_{min} = \dfrac{1}{e^2}$, $|w|_{max} = e^2$ (D) $|w|_{min} = \dfrac{1}{2}$, $|w|_{max} = e$

Paragraph for question nos. 6 to 8

Let A, B, C be three sets of complex numbers as defined below.

$$A = \{z : |z+1| \le 2 + \text{Re}(z)\},\ B = \{z : |z-1| \ge 1\} \text{ and } C = \left\{z : \left|\frac{z-1}{z+1}\right| \ge 1\right\}$$

6. The number of point(s) having integral coordinates in the region $A \cap B \cap C$ is

CN0077

7. The area of region bounded by $A \cap B \cap C$ is $2\sqrt{\lambda}$ then λ is

CN0077

8. The real part of the complex number in the region $A \cap B \cap C$ and having maximum amplitude can be

CN0077

9. Match the List **CN0215**

List - I (Complex number Z)	**List - II** (Principal argument of Z)
(A) $Z = \dfrac{(1+i)^5\left(1+\sqrt{3}i\right)^2}{-2i\left(-\sqrt{3}+i\right)}$.	(P) π
(B) $Z = \sin\dfrac{6\pi}{5} + i\left(1+\cos\dfrac{6\pi}{5}\right)$ is	(Q) $-\dfrac{7\pi}{18}$
(C) $Z = 1 + \cos\left(\dfrac{11\pi}{9}\right) + i\sin\left(\dfrac{11\pi}{9}\right)$	(R) $\dfrac{9\pi}{10}$
(D) $Z = \sin x \sin(x-60)\sin(x+60)$	(S) $-\dfrac{5\pi}{12}$
where $x \in \left(0, \dfrac{\pi}{3}\right)$ and $x \in R$	(T) 0

(A) (A) → (P), (B) → (R), (C) → (S), (D) → (Q)

(B) (A) → (S), (B) → (R), (C) → (Q), (D) → (P)

(C) (A) → (R), (B) → (Q), (C) → (S), (D) → (P)

(D) (A) → (Q), (B) → (S), (C) → (R), (D) → (P)

10. Match the equation in z, in **Column-I** with the corresponding values of arg(z) in **Column-II**.

Column-I (equations in z)		**Column-II** (principal value of arg (z))	
(A)	$z^2 - z + 1 = 0$	(P)	$-2\pi/3$
(B)	$z^2 + z + 1 = 0$	(Q)	$-\pi/3$
(C)	$2z^2 + 1 + i\sqrt{3} = 0$	(R)	$\pi/3$
(D)	$2z^2 + 1 - i\sqrt{3} = 0$	(S)	$2\pi/3$

CN0078

EXERCISE (O-4)

Numerical Grid Type

1. If a and b are positive integer such that $N = (a + ib)^3 - 107i$ is a positive integer then find the value of $\frac{N}{2}$. **CN0216**

2. Let z,w be complex numbers such that $\bar{z} + i\bar{w} = 0$ and arg zw = π. If Re(z) < 0 and principal arg z = $\frac{a\pi}{b}$ then find the value of a + b. (where a & b are co-prime natural numbers) **CN0217**

3. If $x = 9^{1/3}\, 9^{1/9}\, 9^{1/27} \ldots \infty$, $y = 4^{1/3}\, 4^{-1/9}\, 4^{1/27} \ldots \infty$, and $z = \sum_{r=1}^{\infty}(1+i)^{-r}$ and principal argument of P = (x + yz) is $-\tan^{-1}\left(\frac{\sqrt{a}}{b}\right)$ then determine $a^2 + b^2$. (where a & b are co-prime natural numbers) **CN0218**

4. $z_1, z_2 \in c$ and $z_1^2 + z_2^2 \in R$, $z_1(z_1^2 - 3z_2^2) = 2$, $z_2(3z_1^2 - z_2^2) = 11$. If $z_1^2 + z_2^2 = \lambda$ then determine λ^2 **CN0219**

5. Let |z| = 2 and w = $\frac{z+1}{z-1}$ where z, w ∈ C (where C is the set of complex numbers), then find product of maximum and minimum value of |w|. **CN0220**

6. A function 'f' is defined by $f(z) = (4 + i)z^2 + \alpha z + \gamma$ for all complex number z, where α and γ are complex numbers if f(1) and f(i) are both real and the smallest possible values of |α| + |γ| is p then determine p. **CN0221**

7. If z and ω are two non-zero complex numbers such that |zω| = 1, and arg (z) – arg(ω) = $\frac{\pi}{2}$, then find the value of $\frac{5i\bar{z}\omega}{4}$ **CN0222**

8. If $a_1, a_2, a_3 \ldots a_n, A_1, A_2, A_3 \ldots A_n$, k are all real numbers and number of imaginary roots of the equation $\frac{A_1^2}{x - a_1} + \frac{A_2^2}{x - a_2} + \ldots + \frac{A_n^2}{x - a_n} = k$ is α. Then find the value of $\frac{\alpha + 15}{10}$. **CN0223**

9. If a variable circle S touches $S_1 : |z - z_1| = 7$ internally and $S_2 : |z - z_2| = 4$ externally while the curves S_1 & S_2 touch internally to each other, $(z_1 \neq z_2)$. If the eccentricity of the locus of the centre of the curve S is 'e' find the value of e. **CN0224**

10. Area of the region formed by |z| ≤ 4 & $-\frac{\pi}{2} \le \arg z \le \frac{\pi}{3}$ on the Argand diagram is expressed in the form of $\frac{a\pi}{b}$. Then find the value of $\frac{a}{b}$ (where a & b are co-prime natural number) **CN0225**

EXERCISE (JM)

1. Let ω be a complex number such that 2ω + 1 = z where $z = \sqrt{-3}$. If $\begin{vmatrix} 1 & 1 & 1 \\ 1 & -\omega^2-1 & \omega^2 \\ 1 & \omega^2 & \omega^7 \end{vmatrix} = 3k$, then k is equal to :- **[JEE(Main)-2017]**

(1) 1 (2) –z (3) z (4) –1

CN0150

2. If α, β ∈ C are the distinct roots of the equation $x^2 - x + 1 = 0$, then $\alpha^{101} + \beta^{107}$ is equal to- **[JEE(Main)-2018]**

(1) 0 (2) 1 (3) 2 (4) –1

CN0151

3. Let z be a complex number such that |z| + z = 3 + i (where $i = \sqrt{-1}$). Then |z| is equal to :- **[JEE(Main)-2019]**

(1) $\frac{5}{4}$ (2) $\frac{\sqrt{41}}{4}$ (3) $\frac{\sqrt{34}}{3}$ (4) $\frac{5}{3}$

CN0152

4. If $\frac{z-\alpha}{z+\alpha} (\alpha \in R)$ is a purely imaginary number and |z| = 2, then a value of α is : **[JEE(Main)-2019]**

(1) 1 (2) 2 (3) $\sqrt{2}$ (4) $\frac{1}{2}$

CN0153

5. Let Z_1 and Z_2 be two complex numbers satisfying $|Z_1| = 9$ and $|Z_2-3-4i|=4$. Then the minimum value of $|Z_1-Z_2|$ is : **[JEE(Main)-2019]**

(1) 0 (2) 1 (3) $\sqrt{2}$ (4) 2

CN0154

6. If $z = \frac{\sqrt{3}}{2} + \frac{i}{2} \left(i = \sqrt{-1}\right)$, then $\left(1 + iz + z^5 + iz^8\right)^9$ is equal to **[JEE(Main)-2019]**

(1) –1 (2) 1 (3) 0 (4) $(-1+2i)^9$

CN0155

7. Let z∈C be such that |z| < 1. If $\omega = \frac{5+3z}{5(1-z)}$, then :- **[JEE(Main)-2019]**

(1) 5Im(ω) < 1 (2) 4Im(ω) > 5 (3) 5Re(ω) > 1 (4) 5Re(ω) > 4

CN0156

8. If z and w are two complex numbers such that |zw| = 1 and arg(z) – arg(w) = $\frac{\pi}{2}$, then :

[JEE(Main)-2019]

(1) $\bar{z}w = i$ (2) $\bar{z}w = -i$ (3) $z\bar{w} = \frac{1-i}{\sqrt{2}}$ (4) $z\bar{w} = \frac{-1+i}{\sqrt{2}}$

CN0157

9. The equation |z–i| = |z–1|, i = $\sqrt{-1}$, represents: **[JEE(Main)-2019]**

(1) the line through the origin with slope –1. (2) a circle of radius 1.

(3) a circle of radius $\frac{1}{2}$. (4) the line through the origin with slope 1.

CN0158

10. If $\mathrm{Re}\left(\frac{z-1}{2z+i}\right)$ = 1, where z = x + iy, then the point (x,y) lies on a : **[JEE(Main)-2020]**

(1) circle whose centre is at $\left(-\frac{1}{2},-\frac{3}{2}\right)$ (2) circle whose diameter is $\frac{\sqrt{5}}{2}$

(3) straight line whose slope is $\frac{3}{2}$ (4) straight line whose slope is $-\frac{2}{3}$

CN0159

11. Let $\alpha = \frac{-1+i\sqrt{3}}{2}$. If $a = (1+\alpha)\sum_{k=0}^{100}\alpha^{2k}$ and $b = \sum_{k=0}^{100}\alpha^{3k}$, then a and b are the roots of the quadratic equation: **[JEE(Main)-2020]**

(1) $x^2 - 102x + 101 = 0$ (2) $x^2 + 101x + 100 = 0$

(3) $x^2 - 101x + 100 = 0$ (4) $x^2 + 102x + 101 = 0$

CN0160

12. If the equation, $x^2 + bx + 45 = 0$ $(b \in R)$ has conjugate complex roots and they satisfy $|z+1| = 2\sqrt{10}$, then **[JEE(Main)-2020]**

(1) $b^2 - b = 42$ (2) $b^2 + b = 12$ (3) $b^2 + b = 72$ (4) $b^2 - b = 30$

CN0161

13. If z be a complex number satisfying |Re(z)| + |Im(z)| = 4, then |z| cannot be **[JEE(Main)-2020]**

(1) $\sqrt{\frac{17}{2}}$ (2) $\sqrt{10}$ (3) $\sqrt{8}$ (4) $\sqrt{7}$

CN0162

14. Let z be complex number such that $\left|\frac{z-i}{z+2i}\right|=1$ and $|z|=\frac{5}{2}$. Then the value of $|z+3i|$ is :

[JEE(Main)-2020]

(1) $\sqrt{10}$ (2) $2\sqrt{3}$ (3) $\frac{7}{2}$ (4) $\frac{15}{4}$ **CN0163**

15. If $\left(\sqrt{3}+i\right)^{100}=2^{99}(p+iq)$, then p and q are roots of the equation : **[JEE(Main)-2021]**

(1) $x^2-\left(\sqrt{3}-1\right)x-\sqrt{3}=0$ (2) $x^2+\left(\sqrt{3}-1\right)x+\sqrt{3}=0$ **CN0226**

(3) $x^2+\left(\sqrt{3}-1\right)x-\sqrt{3}=0$ (4) $x^2-\left(\sqrt{3}-1\right)x+\sqrt{3}=0$

16. The least value of |z| where z is complex number which satisfies the inequality **CN0227**

$\left(\frac{(|z|+3)(|z|-1)}{||z|+1|}\log_e 2\right)\geq\log_{\sqrt{2}}\left|5\sqrt{7}+9i\right|$, $i=\sqrt{-1}$, is equal to : **[JEE(Main)-2021]**

(1) 3 (2) $\sqrt{5}$ (3) 2 (4) 8

17. Let z_1, z_2 be the roots of the equation $z^2 + az + 12 = 0$ and z_1,z_2 form an equilateral triangle with origin. Then, the value of |a| is **[JEE(Main)-2021]**

CN0228

18. Let a circle C in complex plane pass through the points $z_1 = 3 + 4i$, $z_2 = 4 + 3i$ and $z_3 = 5i$. If $z(\neq z_1)$ is a point on C such that the line through z and z_1 is perpendicular to the line through z_2 and z_3, then arg(z) is equal to : **[JEE(Main)-2022]**

(A) $\tan^{-1}\left(\frac{2}{\sqrt{5}}\right)-\pi$ (B) $\tan^{-1}\left(\frac{24}{7}\right)-\pi$ (C) $\tan^{-1}(3)-\pi$ (D) $\tan^{-1}\left(\frac{3}{4}\right)-\pi$

CN0229

19. Let α be a root of the equation $1 + x^2 + x^4 = 0$. Then the value of $\alpha^{1011} + \alpha^{2022} - \alpha^{3033}$ is equal to :

[JEE(Main)-2022]

(A) 1 (B) α (C) 1 + α (D) 1 + 2α **CN0230**

20. Let z = a + ib, b ≠ 0 be complex numbers satisfying $z^2=\bar{z}\cdot 2^{1-|z|}$. Then the least value of n ∈ N, such that $z^n = (z+1)^n$, is equal to _____ **[JEE(Main)-2022]**

CN0231

EXERCISE (JA)

1. Let complex numbers α and $\frac{1}{\bar{\alpha}}$ lie on circles $(x - x_0)^2 + (y - y_0)^2 = r^2$ and $(x - x_0)^2 + (y - y_0)^2 = 4r^2$ respectively. If $z_0 = x_0 + iy_0$ satisfies the equation $2|z_0|^2 = r^2 + 2$, then $|\alpha| =$

[JEE(Advanced) 2013, 3M]

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{2}$ (C) $\frac{1}{\sqrt{7}}$ (D) $\frac{1}{3}$

CN0173

2. Let ω be a complex cube root of unity with $\omega \neq 1$ and $P = [p_{ij}]$ be a $n \times n$ matrix with $p_{ij} = \omega^{i+j}$. Then $P^2 \neq 0$, when $n =$ **[JEE(Advanced) 2013, 4, (–1)]**

(A) 57 (B) 55 (C) 58 (D) 56

CN0174

3. Let $w = \frac{\sqrt{3}+i}{2}$ and $P = \{w^n : n = 1, 2, 3, ...\}$. Further $H_1 = \left\{z \in C : \text{Re}\, z > \frac{1}{2}\right\}$ and $H_2 = \left\{z \in C : \text{Re}\, z < \frac{-1}{2}\right\}$, where C is the set of all complex numbers. If $z_1 \in P \cap H_1, z_2 \in P \cap H_2$ and O represents the origin, then $\angle z_1Oz_2 =$ **[JEE-Advanced 2013, 4, (–1)]**

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{6}$ (C) $\frac{2\pi}{3}$ (D) $\frac{5\pi}{6}$ **CN0175**

Paragraph for Question 4 and 5

Let $S = S_1 \cap S_2 \cap S_3$, where $S_1 = \{z \in C : |z| < 4\}$, $S_2 = \left\{z \in C : \text{Im}\left[\frac{z-1+\sqrt{3}i}{1-\sqrt{3}i}\right] > 0\right\}$ and $S_3 = \{z \in C : \text{Re}\, z > 0\}$.

4. $\min\limits_{z \in S} |1-3i-z| =$ **[JEE(Advanced) 2013, 3, (–1)]**

(A) $\frac{2-\sqrt{3}}{2}$ (B) $\frac{2+\sqrt{3}}{2}$ (C) $\frac{3-\sqrt{3}}{2}$ (D) $\frac{3+\sqrt{3}}{2}$

CN0176

5. Area of S = **[JEE(Advanced) 2013, 3, (–1)]**

(A) $\dfrac{10\pi}{3}$ (B) $\dfrac{20\pi}{3}$ (C) $\dfrac{16\pi}{3}$ (D) $\dfrac{32\pi}{3}$

CN0176

6. The quadratic equation $p(x) = 0$ with real coefficients has purely imaginary roots. Then the equation $p(p(x)) = 0$ has **[JEE(Advanced) 2014, 3(–1)]**

(A) only purely imaginary roots

(B) all real roots

(C) two real and two purely imaginary roots

(D) neither real nor purely imaginary roots.

CN0177

7. Let $z_k = \cos\left(\dfrac{2k\pi}{10}\right) + i\sin\left(\dfrac{2k\pi}{10}\right)$; k = 1, 2, ... 9.

	List-I		**List-II**
P.	For each z_k there exists a z_j such that $z_k \cdot z_j = 1$	1.	True
Q.	There exists a $k \in \{1, 2, ..., 9\}$ such that $z_1 \cdot z = z_k$ has no solution z in the set of complex numbers.	2.	False
R.	$\dfrac{\|1-z_1\|\|1-z_2\|...\|1-z_9\|}{10}$ equals	3.	1
S.	$1 - \sum_{k=1}^{9}\cos\left(\dfrac{2k\pi}{10}\right)$ equals	4.	2

Codes :

	P	Q	R	S
(A)	1	2	4	3
(B)	2	1	3	4
(C)	1	2	3	4
(D)	2	1	4	3

[JEE(Advanced) 2014, 3(–1)]

CN0178

8. **Column-I** **Column-II**

(A) In $\mathbb{R}^2$, if the magnitude of the projection vector of the vector $\alpha\hat{i}+\beta\hat{j}$ on $\sqrt{3}\hat{i}+\hat{j}$ is $\sqrt{3}$ and if $\alpha = 2+\sqrt{3}\beta$, then possible value(s) of $|\alpha|$ is (are) — (P) 1

(B) Let a and b be real numbers such that the function $f(x)=\begin{cases}-3ax^2-2, & x<1\\ bx+a^2, & x\ge 1\end{cases}$ is differentiable for all $x \in \mathbb{R}$. Then possible value(s) of a is (are) — (Q) 2

(C) Let $\omega \neq 1$ be a complex cube root of unity. If $(3-3\omega+2\omega^2)^{4n+3} + (2+3\omega-3\omega^2)^{4n+3} + (-3+2\omega+3\omega^2)^{4n+3} = 0$ then possible value(s) of n is (are) — (R) 3

(D) Let the harmonic mean of two positive real number a and b be 4, If q is a positive real number such that a, 5, q, b is an arithmetic progression, then the value(s) of $|q-a|$ is (are) — (S) 4

(T) 5

[JEE 2015, 8(Each 2M, –1M)]

CN0179

9. For any integer k, let $\alpha_k = \cos\left(\frac{k\pi}{7}\right)+i\sin\left(\frac{k\pi}{7}\right)$, where $i=\sqrt{-1}$. The value of the expression

$$\frac{\sum_{k=1}^{12}|\alpha_{k+1}-\alpha_k|}{\sum_{k=1}^{3}|\alpha_{4k-1}-\alpha_{4k-2}|}$$ is

[JEE 2015, 4M, –0M]

CN0180

10. Let $z = \frac{-1+\sqrt{3}i}{2}$, where $i = \sqrt{-1}$, and $r, s \in \{1, 2, 3\}$. Let $P = \begin{bmatrix}(-z)^r & z^{2s}\\ z^{2s} & z^r\end{bmatrix}$ and I be the identity matrix of order 2. Then the total number of ordered pairs (r, s) for which $P^2 = -I$ is

[JEE(Advanced)-2016, 3(0)]

CN0181

11. Let a,b $\in \mathbb{R}$ and $a^2 + b^2 \neq 0$. Suppose $S=\left\{z\in\mathbb{C}:z=\frac{1}{a+ibt},t\in\mathbb{R},t\neq 0\right\}$, where $i=\sqrt{-1}$.

If z = x + iy and z $\in$ S, then (x,y) lies on

(A) the circle with radius $\frac{1}{2a}$ and centre $\left(\frac{1}{2a},0\right)$ for a > 0, b $\neq$ 0

(B) the circle with radius $-\frac{1}{2a}$ and centre $\left(-\frac{1}{2a},0\right)$ for a < 0, b $\neq$ 0

(C) the x-axis for a $\neq$ 0, b = 0

(D) the y-axis for a = 0, b $\neq$ 0

[JEE(Advanced)-2016, 4(–2)]

CN0182

12. Let a, b, x and y be real numbers such that a – b = 1 and y $\neq$ 0. If the complex number z = x + iy satisfies $\operatorname{Im}\left(\frac{az+b}{z+1}\right)=y$, then which of the following is(are) possible value(s) of x ?

[JEE(Advanced)-2017, 4(–2)]

(A) $-1-\sqrt{1-y^2}$ (B) $1+\sqrt{1+y^2}$ (C) $1-\sqrt{1+y^2}$ (D) $-1+\sqrt{1-y^2}$

CN0183

13. For a non-zero complex number z, let arg(z) denotes the principal argument with $-\pi<\arg(z)\leq\pi$. Then, which of the following statement(s) is (are) **FALSE** ? **[JEE(Advanced)-2018, 4(–2)]**

(A) $\arg(-1-i)=\frac{\pi}{4}$, where $i=\sqrt{-1}$

(B) The function $f:\mathbb{R}\to(-\pi,\pi]$, defined by $f(t)=\arg(-1+it)$ for all $t\in\mathbb{R}$, is continuous at all points of $\mathbb{R}$, where $i=\sqrt{-1}$

(C) For any two non-zero complex numbers z_1 and z_2, $\arg\left(\frac{z_1}{z_2}\right)-\arg(z_1)+\arg(z_2)$ is an integer multiple of 2π

(D) For any three given distinct complex numbers z_1, z_2 and z_3, the locus of the point z satisfying the condition $\arg\left(\frac{(z-z_1)(z_2-z_3)}{(z-z_3)(z_2-z_1)}\right)=\pi$, lies on a straight line

CN0184

14. Let s, t, r be the non-zero complex numbers and L be the set of solutions $z = x + iy \ (x, y \in \mathbb{R}, i = \sqrt{-1})$ of the equation $sz + t\bar{z} + r = 0$, where $\bar{z} = x - iy$. Then, which of the following statement(s) is (are) TRUE ? **[JEE(Advanced)-2018, 4(–2)]**

(A) If L has exactly one element, then $|s| \neq |t|$

(B) If $|s| = |t|$, then L has infinitely many elements

(C) The number of elements in $L \cap \{z : |z - 1 + i| = 5\}$ is at most 2

(D) If L has more than one element, then L has infinitely many elements

CN0185

15. Let S be the set of all complex numbers z satisfying $|z - 2 + i| \geq \sqrt{5}$. If the complex number z_0 is such that $\frac{1}{|z_0 - 1|}$ is the maximum of the set $\left\{\frac{1}{|z-1|} : z \in S\right\}$, then the principal argument of $\frac{4 - z_0 - \bar{z}_0}{z_0 - \bar{z}_0 + 2i}$ is **[JEE(Advanced)-2019, 3(–1)]**

(1) $\frac{\pi}{4}$ (2) $-\frac{\pi}{2}$ (3) $\frac{3\pi}{4}$ (4) $\frac{\pi}{2}$

CN0186

16. Let $\omega \neq 1$ be a cube root of unity. Then the minimum of the set $\{|a + b\omega + c\omega^2|^2 : a, b, c$ distinct non-zero integers$\}$ equals ____ **[JEE(Advanced)-2019, 3(0)]**

CN0187

17. Let S be the set of all complex numbers z satisfying $|z^2 + z + 1| = 1$. Then which of the following statements is/are TRUE ? **[JEE(Advanced)-2020]**

(A) $\left|z + \frac{1}{2}\right| \leq \frac{1}{2}$ for all $z \in S$

(B) $|z| \leq 2$ for all $z \in S$

(C) $\left|z + \frac{1}{2}\right| \geq \frac{1}{2}$ for all $z \in S$

(D) The set S has exactly four elements

CN0193

18. For a complex number z, let Re(z) denote the real part of z. Let S be the set of all complex numbers z satisfying $z^4 - |z|^4 = 4iz^2$, where $i = \sqrt{-1}$. Then the minimum possible value of $|z_1 - z_2|^2$, where $z_1, z_2 \in S$ with $\text{Re}(z_1) > 0$ and $\text{Re}(z_2) < 0$, is _____ **[JEE(Advanced)-2020]**

CN0194

19. Let $\theta_1, \theta_2, ..., \theta_{10}$ be positive valued angles (in radian) such that $\theta_1 + \theta_2 + ... + \theta_{10} = 2\pi$. Define the complex numbers $z_1 = e^{i\theta_1}$, $z_k = z_{k-1}e^{i\theta_k}$ for k = 2, 3, ..., 10, where $i = \sqrt{-1}$. Consider the statements P and Q given below : **[JEE(Advanced)-2021]**

P : $|z_2 - z_1| + |z_3 - z_2| + ... + |z_{10} - z_9| + |z_1 - z_{10}| \le 2\pi$

Q : $|z_2^2 - z_1^2| + |z_3^2 - z_2^2| + ... + |z_{10}^2 - z_9^2| + |z_1^2 - z_{10}^2| \le 4\pi$

Then,

(A) P is **TRUE** and Q is **FALSE**

(B) Q is **TRUE** and P is **FALSE**

(C) both P and Q are **TRUE**

(D) both P and Q are **FALSE**

CN0195

20. For any complex number w = c + id, let $\arg(w) \in (-\pi, \pi]$, where $i = \sqrt{-1}$. Let α and β be real numbers such that for all complex numbers z = x + iy satisfying $\arg\left(\frac{z+\alpha}{z+\beta}\right) = \frac{\pi}{4}$, the ordered pair (x,y) lies on the circle $x^2 + y^2 + 5x - 3y + 4 = 0$. Then which of the following statements is (are) **TRUE** ?

(A) $\alpha = -1$

(B) $\alpha\beta = 4$

(C) $\alpha\beta = -4$

(D) $\beta = 4$

[JEE(Advanced)-2021]

CN0196

21. Let z be a complex number with non-zero imaginary part. If $\frac{2+3z+4z^2}{2-3z+4z^2}$ is a real number, then the value of $|z|^2$ is ____________. **[JEE(Advanced)-2022]**

CN0232

22. Let $\bar{z}$ denote the complex conjugate of a complex number z and let $i = \sqrt{-1}$. In the set of complex numbers, the number of distinct roots of the equation $\bar{z} - z^2 = i(\bar{z} + z^2)$ is ____________.

[JEE(Advanced)-2022]

CN0233

23. Let $\bar{z}$ denote the complex conjugate of a complex number z. If z is a non-zero complex number for which both real and imaginary parts of $(\bar{z})^2 + \frac{1}{z^2}$ are integers, then which of the following is/are possible value(s) of |z| ? **[JEE(Advanced)-2022]**

(A) $\left(\frac{43+3\sqrt{205}}{2}\right)^{\frac{1}{4}}$

(B) $\left(\frac{7+\sqrt{33}}{4}\right)^{\frac{1}{4}}$

(C) $\left(\frac{9+\sqrt{65}}{4}\right)^{\frac{1}{4}}$

(D) $\left(\frac{7+\sqrt{13}}{6}\right)^{\frac{1}{4}}$

CN0234

24. Let $A=\left\{\frac{1967+1686i\sin\theta}{7-3i\cos\theta}:\theta\in\mathbb{R}\right\}$. If A contains exactly one positive integer n, then the value of n is **[JEE(Advanced)-2023]**

CN0235

25. Let z be complex number satisfying $|z|^3+2z^2+4\bar{z}-8=0$, where $\bar{z}$ denotes the complex conjugate of z. Let the imaginary part of z be nonzero. **[JEE(Advanced)-2023]**

Match each entry in **List-I** to the correct entries in **List-II**. **CN0236**

List-I		**List-II**	
(P)	$\lvert z\rvert^2$ is equal to	(1)	12
(Q)	$\lvert z-\bar{z}\rvert^2$ is equal to	(2)	4
(R)	$\lvert z\rvert^2+\lvert z+\bar{z}\rvert^2$ is equal to	(3)	8
(S)	$\lvert z+1\rvert^2$ is equal to	(4)	10
		(5)	7

The correct option is :

(A) (P) → (1) (Q) → (3) (R) → (5) (S) → (4)

(B) (P) → (2) (Q) → (1) (R) → (3) (S) → (5)

(C) (P) → (2) (Q) → (4) (R) → (5) (S) → (1)

(D) (P) → (2) (Q) → (3) (R) → (5) (S) → (4)

26. Let $A_1, A_2, A_3, \ldots\ldots, A_8$ be the vertices of a regular octagon that lie on a circle of radius 2. Let P be a point on the circle and let PA_i denote the distance between the points P and A_i for $i = 1,2,\ldots,8$. If P varies over the circle, then the maximum value of the product $PA_1\cdot PA_2\cdot\ldots\ldots\cdot PA_8$, is

[JEE(Advanced)-2023]

CN0237

ANSWER KEY

Do yourself - 1

1. $n = 4$ **2.** 0 **3.** $\begin{cases} (1,\ 0) & \text{for} \quad n = 4k \\ (1,\ 1) & \text{for} \quad n = 4k+1 \\ (0,\ 1) & \text{for} \quad n = 4k+2 \\ (0,\ 0) & \text{for} \quad n = 4k+3 \end{cases}$

Do yourself - 2

1. $-17 + 24i$ **3.** $\pm(1 - 4i)$

4. **(a)** $\frac{7}{25} + \frac{24}{25}i$; **(b)** $\frac{21}{5} - \frac{12}{5}i$; **(c)** $3 + 4i$; **(d)** $\frac{22}{5}i$;

5. **(a)** $x = 1, y = 2$; **(b)** $(2,9)$; **(c)** $(-2, 2)$ or $\left(-\frac{2}{3}, -\frac{2}{3}\right)$

6. **(a)** $\pm(5 + 4i)$; **(b)** $\pm(5 - 6i)$; **(c)** $\pm 5(1 + i)$

7. **(a)** -160; **(b)** $-(77 + 108 i)$

8. **(a)** $-i, -2i$ **(b)** $\frac{3 - 5i}{2}$ or $-\frac{1+i}{2}$

10. B **11.** A **12.** A

13. **(a)** 2, **(b)** $-11/2$ **14.** **(a)** $z = (2 + i)$ or $(1 - 3i)$; **(b)** $z = 1$ or $2 - i$

15. $\frac{iz}{2} + \frac{1}{2} + i$ **16.** 26 **17.** 51

Do yourself - 3

1. **(i)** $|z| = 4$; $amp(z) = \frac{2\pi}{3}$ **(ii)** $|z| = 2$; $amp(z) = -\frac{5\pi}{6}$ **(iii)** $|z| = 2$; $amp(z) = -\frac{\pi}{2}$

(iv) $|z| = \frac{1}{\sqrt{2}}$; $amp(z) = \frac{3\pi}{4}$ **(v)** $|z| = 2$; $amp(z) = \frac{\pi}{3}$

2. 15 **3.** $x = 1, y = -4$ or $x = -1, y = -4$ **4.** $[(-2, 2)\ ;\ (-2, -2)]$

Do yourself - 4

1. 13 units

2. locus is a circle on complex plane with center at (2,3) and radius 1 unit.

3. C

4. **(a)** on a circle of radius $\sqrt{7}$ with centre $(-1, 2)$; **(b)** on a unit circle with centre at origin **(c)** on a circle with centre $(-15/4, 0)$ & radius $9/4$; **(d)** a straight line.

5. A **6.** $a = b = 2 - \sqrt{3}$ **7.** $z_3 = \sqrt{3}(1 - i)$ and $z'_3 = \sqrt{3}(-1 + i)$

Do yourself - 5

1. **(i)** $2\sqrt{2}\left(\cos\frac{3\pi}{4}+i\sin\frac{3\pi}{4}\right)$; $2\sqrt{2}e^{i\left(\frac{3\pi}{4}\right)}$ **(ii)** $2\left(\cos\frac{4\pi}{3}+i\sin\frac{4\pi}{3}\right)$; $2e^{i\left(\frac{4\pi}{3}\right)}$

(iii) $\sqrt{2}\left(\cos\frac{3\pi}{4}+i\sin\frac{3\pi}{4}\right)$; $\sqrt{2}e^{i\left(\frac{3\pi}{4}\right)}$ **(iv)** $2\sin\left(\frac{\theta}{2}\right)\left(\cos\left(\frac{\pi}{2}-\frac{\theta}{2}\right)+i\sin\left(\frac{\pi}{2}-\frac{\theta}{2}\right)\right)$; $2\sin\left(\frac{\theta}{2}\right)e^{i\left(\frac{\pi}{2}-\frac{\theta}{2}\right)}$

2. B **3.** B

Do yourself - 6

1. C **2.** D **4.** $\frac{x^2}{64}+\frac{y^2}{48}=1$ **5.** A **6.** A

7. D **8.** B **9.** A **10.** B,C,D

11. **(a)** $\frac{\sqrt{3}}{2}-\frac{i}{2}, -\frac{\sqrt{3}}{2}-\frac{i}{2}, i$ **12.** D **13.** D

14. **(i)** Modulus = 6, Arg $= 2k\pi+\frac{5\pi}{18}$ $(k \in I)$, Principal Arg $= \frac{5\pi}{18}$

(ii) Modulus = 2, Arg $= 2k\pi+\frac{7\pi}{6}$ $(k \in I)$, Principal Arg $= -\frac{5\pi}{6}$

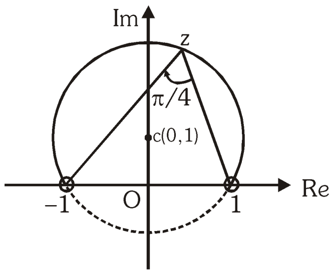
(iii) Modulus $= \frac{\sqrt{5}}{6}$, Arg $= 2k\pi - \tan^{-1}2$ $(k \in I)$, Principal Arg $= -\tan^{-1}2$

15. **(a)** 1, **(b)** 2 **(c)** 30 **17.** 673

Do yourself - 7

1. $1+(2+2\sqrt{2})i$ **2.** 0 **3.** $z_2 = z_3 + i(z_1 - z_3)$ **4.** B

5. Locus is all the points on the major arc of circle as shown excluding points 1 & –1.



6. B **7.** A **8.** A **9.** C

Do yourself - 8

1. C **2.** A,D **3.** 2^{1000} **4.** 1 **7.** A

8. D **9.** C

Do yourself - 9

1. D **2.** B **3.** C **4.** A **5.** C

6. 6 **7.** $\left[\frac{n(n+1)}{2}\right]^2 - n$

Do yourself - 10

1. A **2.** A **4.** A **6.** D **7.** B

8. D **9.** (a) $-\frac{7}{2}$, (b) zero **10.** 4 **11.** $48(1-i)$

Do yourself - 11

3. no point of intersection **7.** A **8.** D **9.** $-\frac{3}{2}+\frac{3\sqrt{3}}{2}i$

10. **(a)** $\pi - 2$, **(b)** -4

11. **(a)** The region between the concentric circles with centre at (0 , 2) & radii 1 & 3 units

(b) region outside or on the circle with centre $\frac{1}{2}+2i$ and radius $\frac{1}{2}$.

(c) semicircle (in the 1st & 4th quadrant) $x^2 + y^2 = 1$

(d) a ray emanating from the point $(3 + 4i)$ directed away from the origin & having equation $\sqrt{3}x - y + 4 - 3\sqrt{3} = 0$

12. 41

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	C	B	C	D	B	C	B	D	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	C	A	B	B	B	A	C	C	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	A	C	A	B	B	D	D	A	D

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C,D	A,B	B,C,D	A,B,C,D	B,D	A,B,C,D	B,C,D	B,C	A,B,C,D	A,B,C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B,D	B,C	B,D	A,B,C	A,B,C,D	A,B,D	A,B	A,B,C,D	A,B	A,B,C
Que.	21	22	23	24	25					
Ans.	A,B	A,C	B,C,D	A,B,C,D	A,C,D					

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8	9
Ans.	B	C	C	A,B	B,D	5.00	3.00	–1.50	B
Q.10	A	B	C	D					
	Q,R	P,S	Q,S	P,R					

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	99	7	13	25	1	1.41 or 1.42	1.25	1.50	0.27 or 0.28	6.66 or 6.67

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	2	4	2	1	1	3	2	4	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	4	4	3	1	1	6	B	A	6

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7			
Ans.	C	B,C,D	C,D	C	B	D	C			
Q.8	A	B	C	D	9	10	11	12	13	14
	P,Q	P,Q	P,Q,S,T	Q,T	4	1	A,C,D	A,D	A,B,D	A,C,D
Que.	15	16	17	18	19	20	21	22	23	24
Ans.	2	3.00	B,C	8	C	B,D	0.50	4.00	A	281
Que.	25	26								
Ans.	B	512								